

ActInf GuestStream #015.1: Bobby Azarian, Universal Bayesianism: A New

GuestStream | 2:37:08

Yep. All right. Thanks, Daniel, for having me, everyone at the Active Inference Lab.

Sorry for canceling twice. Life's been kind of crazy with finishing this book that this talk will be about and having a kid. I'm six months old and I got sick like a week ago, like right leading up to like, you know, didn't want to cancel a third time and look crazy. So this is a very rushed presentation. It'll be better if it's more of like a conversation since I haven't even gone through the slides myself once. And it's going to get into a lot of things like immediately, like open up bags of worms, like there's just philosophical issues, things that people won't agree about, like immediately. But yeah, so this is just going to be like a small introduction to the thesis in the book. And yeah, I'll explain more, I guess, after I begin. So yeah, it's called this the integrated evolutionary synthesis, compared to like the modern synthesis, or even the extended evolutionary synthesis. This is integrated because it kind of reconceptualizes evolution, biology and evolution in terms of thermodynamics and information theory or energy flows, energy and information flows takes from cybernetics. So yeah, it's a non reductive theory of everything that includes emergent phenomena like life, mind and civilization. I say that because most theories of everything we're used to are fundamental physics theories that don't have anything to say about life and consciousness. They ignore them. And I don't think those should be called theories of everything because theories of everything should be an explanation for everything. And if you leave the things that we care about most out, then it's not much of a theory of everything. So this is the book I've spent the last two to three years writing obsessively. And it's available for pre-order now. So if you want to check that out,

you can find it with a Google search. Here's the table of contents. So the book is broken into three sections. Part one talks about the origins of life and part two, about evolution, while part three focuses on consciousness and free will and the fate of life in the universe. So these like really big questions, but we're going to focus on part two and give a pretty like a superficial overview of part two.

But you can feel free to ask me any questions about any of the other topics that are related. And I'll try to see what I can answer without going into like all of the information that I would, I would need to explain those things.

So Thomas Huxley, known as Darwin's bulldog,

because he was such a strong advocate for Darwin's theory, said that the question of questions for mankind, the problem which underlies all others is more deeply interesting than any other, is the ascertainment and of the place which man occupies in nature and of his relations to the universe of things. So where do we fit into the grand cosmic scheme? Most scientists in the 20th century, and I would even say now, believe that we are an accident. So Jacques Monod, a French biochemist who won the Nobel Prize for

physiology and medicine, wrote a book called Chance and Necessity, which was really influential. And

he was just as reductionist as you can get. Man at last knows that he is alone in the unfeeling immensity of the universe, out of which he emerged only by chance. Neither, so some of the words are covered by Daniel. So neither is his destiny spelled out nor his duty have been written down. I think there's something wrong with that quote. But so if there are times where I'm trying to read something where the picture's over, I might have to skip. You can move that because it looks different on my side. It looks fine. Thank you. Great. Yeah. Neither his destiny nor his duty have been written down. So that was kind of based on this old idea that the emergence of life was a product of chance assembly. So like this just statistical fluctuation that brought together all the molecules needed to create the first cell. We now know that that's probably not what happened. It's this gradual process of self-organization. Jeremy England has been in the media a lot about his theory of dissipative adaptation. So let's see. So we will see this is kind of like a different point of view. Not everybody was convinced by a monitor. Carl Sagan said the origin of life must be a highly probable affair. As soon as conditions permit, up it pops. So this is an important question. Whether life was improbable or inevitable. So if life is highly improbable, we're likely to be alone in the universe. But if the emergence of life is inevitable, given a certain set of common physical conditions, then it's likely that we have life, at least on planets that are sufficiently Earth-like, specifically have geochemistry like the Earth's, have a star, some like the Earth's, because that's what provides the energy that pushes it far from equilibrium. So we're going to get in to all that stuff pretty soon. Christian Didou, this quote is kind of a response to Stephen Jay Gould, who just really emphasized to both Mano and Stephen Jay Gould, the importance of chance in nature, and that everything is basically, you know, a contingency that came from a chance process. Where Didou says that, you know, you can have this inevitability that some philosophers would see as teleology. But it doesn't mean that this inevitable progress is being driven by a supernatural force. He explains that the natural constraints within which chance operates are such that evolution in the direction of increasing complexity was virtually bound to take place. If given the opportunity, chance does not exclude inevitability. So he was also a Nobel Prize winning biologist, the same category as Mano, but had the complete opposite opinion. And we can talk about that too. Basically, when systems are open systems, and they're pushed far from equilibrium, by a flow of energy, you basically get something like a statistical bias away from what we would expect with isolated systems, closed systems that aren't open to energy will relax the thermodynamic equilibrium. And when a system is being pushed far from equilibrium, it will naturally organize because organization happens under the flow of energy. And we're going to see exactly why it's a Darwinian process. But basically, you get inevitability if you have the right ingredients. So we'll talk about to kind of frame this, we already kind of mentioned those two worldviews. So the first worldview would be the reductionist worldview. And I should be clear that reductionism is also it's a method and an ideology. And we shouldn't confuse those two things. And Daniel, I guess, just so I'm, you know, know where I'm at, maybe when you when I've gone 30 minutes, you can let me know.

Daniel Kahneman Okay, I'll hoot.

Daniel Kahneman Okay, great. So reductionism, the method, basically says that reality can be

understood by breaking down all physical phenomena into their simplest parts so that we may observe the basic behavior of the fundamental constituents of nature. So I'm just gonna plug in this charger, I might need to do that in a bit. But basically, the method is how science works. If we want to understand emergence, we also have to understand how the components that systems are made of work in isolation. So it's an essential part of the story, but it's not the whole part, we'll see. So the method inspired an ideology and ideology is, is, is just that it's, it's an ideology, and it's rigid, and basically, the method doesn't really imply that the ideology is true. So when I'm attacking reductionism, I'm not attacking the method, I'm attacking the worldview. And so that worldview, we can say that, basically, so hard determinism is, is part of the reductionist worldview. So thought experiment that some people are familiar with, I'm sure, Laplace's demon, but basically, Laplace was French mathematician that took Newton's kind of model of reality, and applied it to everything. Newton himself was kind of a mystic, when it comes to life, he didn't really think that physics apply to organisms, or at least to humans. But Laplace really, you know, came up with the idea that like, basically, the evolution of the universe is just this like, clockwork universe, where it's like, a machine with cogs that are just turning. So there are no, there's no room for agency and free will, in that picture. When Boltzmann came along, in the 19th century, theory and came up with statistical mechanics, the clockwork universe, kind of became the billiard ball universe, but it's just the idea that everything in the universe is made of atoms, and that these atoms are interacting, and that's the whole picture. So the ideology says we're nothing more than our atoms. So we don't make decisions. Consciousness is just an epiphenomenon, and it feels to us like we are, but actually, this trajectory was determined, basically, not basically from the moment of the Big Bang. And there's kind of no freedom in anything, and no future, other than what was determined from the beginning. So the second law of thermodynamics also sort of shaped the reductionist world abuse as the universe is becoming increasingly disordered. There's actually kind of two different interpretations of the second law. The first one kind of emerged from the work of Carnot and Clausius, the kind of original thermodynamics work where entropy was about a measure of energy that can't be used for work. So a kind of measure of like useless energy, where later with Boltzmann, he tried to put the second law on like a statistical basis based on like atomic theory. So basically, people applied Boltzmann statistical interpretation of the second law to the universe and came up with this notion that the universe is becoming increasingly disordered, which isn't great for life because it means that life is transient and likely cosmically insignificant. And this worldview is associated with materialism, which doesn't recognize consciousness is real certainly doesn't recognize it as having causal power. And historically, it often ignored, apparently, immaterial phenomena like energy and information, mental processes. I am just going to grab my charger. So Sounds good.

This doesn't die, but I will be right back.

Take your time.

So those who are watching live, please feel free to leave a comment or a question. I'm writing some things down.

I'm writing some things down. We'll look forward to the return of electrical power to Bobby.

All right, my first live stream, I will be more prepared. I return to part two. That's something else I've told Daniel is that this is kind of like part one that I'd like to come back and talk about the ideas that come after. So

It will have 15.2. It'll be awesome. It's an infinite sequence. It's the sequence is always yours.

Great. Okay. So reductionist worldview says that all life forms, including humans are nothing more than collections of atoms obligatorily following fixed and arbitrary mechanical trajectories determined solely by math and not by mind. So this worldview, if you convince someone that they don't have free will, because you tell them about the reductionist worldview.

There was actually a study where they did this and they read participants passage from Francis Crick explaining that it appears that we don't have any free will.

The participants actually on subsequent tests were more likely to cheat on the test because they felt like they didn't have any personal responsibility. They weren't making the decisions. Therefore, they could cheat. That's kind of funny because they are actually you can see that like, by what they're like, being told there they are changing their behavior. So it is having some kind of effect. If we do in fact have agency or free will, and we're kind of talking about agency, we don't really get into free will, but those are related topics. But if that's not true, then that's really bad that the majority of scientists are telling people we don't have free will because the people that believe this, it's having negative effects on, for example, mental health. Other studies have shown that belief in no free will can lead to depression and belief in free will can have all these positive effects.

So, of course, as soon as I start talking about free will, I'm sure a lot of people have objections, might think that there's no way that we can have room for free will in a physical worldview. But that's exactly what we're going to begin to talk about today, the causal power of information. And that basically organisms are agents that are cybernetic control units. So as controllers, we do have agency. So in the belief that we're accidents of nature, rather than natural manifestations of physical laws, I'm sure also impacts our sense of well being just so you know, I'm not making this up. I can I'll give you some quotes from really famous physicists, who are kind of militant reductionists.

So Brian Greene says, I think it's very important to face up to the truth of reality, which is in fact that life and consciousness is a fleeting phenomenon on the entire cosmological timeline.

Brian Greene just came out with a book, and he's like, going on interviews just saying this. And it's really interesting, because some other famous physicists don't agree at all. And he's stating it like it's complete facts. So for example, David Deutsch would not agree with this. And we'll get into that. Sabine Hossenfelder is another person. I know some of the some of my friends in the community, who are these people, you know, working on emergence and causation stuff, get bothered by her statements, because she is also very ultra reductionistic. So I wish people would stop

insisting they have free will. It's terribly annoying insisting that free will exists is bad science, like insisting that horoscopes tell you something about the future. It's not compatible with our knowledge about nature. So I'm saying that's wrong. We won't get into free will that much. But if you have questions at the end, we will talk about how information adaptive information gets built up by evolution in adaptive systems, complex adaptive systems. And you will see how these systems actually do control their own future. So Sam Harris is another anti free will person says, you're not controlling the storm, and you are not lost in it, you are the storm. So it's kind of funny, because he's using a metaphor here. But what this story kind of reveals, is that yes, we are a storm, that's right, we are a dissipative structure, like a storm emerges to dissipate an energy gradient. And we're not really going to talk about that. That's what the first part of the book is about the origins part. But I'm just pointing it out, because it's interesting, because we are like a storm in that aspect. But we also differ from a storm, because a storm does not have agency, a storm does not have the ability to seek out new energy gradients, it's just kind of being pushed around by these gradients in nature where organisms are actually controllers. And so you are storm, but you do have control over the storm.

So these people are kind of have the opposite view. So Giulio Tononi, creator of integrated information theory says there is true free will, a great lecture that just popped up, I think, months ago, where he gives a two hour talk on basically integrate information and how it shows that systems do have causal power and in a way, have free will. So depending on your definition of free will.

We do have free will, according to his model. David Deutsch said in an interview with John Horgan, just a year ago or so, I'm sure we have free will. Christophe Koch says explains how you know, there are a lot of experiments, the Libet experiments, which we will talk about next time, that basically, some people interpret it as not having free will, but actually, we will see later, when you deliberate, that's actually an act of free will, these choices where we're actually where you bring your entire conscious being to that question and try to analyze it under all the various conditions. And actually, there was a study Koch did that kind of showed that the Libet study was kind of interpreted wrong. And for anyone who understands that stuff, the readiness potential that was supposed to be that the proof that there was no free will disappeared when the decisions involved deliberation.

So yeah, that's kind of meandering. But I'm sure a lot of people are interested in this bigger picture of why this emergence paradigm is so radically different from the reductionist paradigm. It kind of puts information front and center and systems that process information, complex adaptive systems are computational systems. And these systems in the with these systems information actually does have causal power, we're going to explain what we mean by that. So paradigm of emergence, the collective says the collective behavior of interacting parts, not simply how they function in isolation is key to understanding the emergence and evolution of all the fascinating organisms and ecosystems that make up the biosphere. So it's really not about just like particles, it's about the patterns that emerge when these systems find these certain configurations through basically Darwinian mechanisms. And we'll see that self organization to has a Darwinian dynamic. Each emergence in the cosmic self organization process is brought about by higher order phase transition that moves life ever further away from a state of

thermodynamic equilibrium and total disorder. I'm assuming the audience knows that what thermodynamic equilibrium is, it's basically if you have, for example, Boltzmann took a model of an ideal gas and explain that for example, if the gas molecules are bunched up in the corner, naturally, it's going, the gas is going to

spread out until there's no pattern. And basically, it's like a completely mixed up state.

So equilibrium is associated with death and disorder, because systems naturally tend toward this more more mixed up, unpatterned, completely chaotic state. Of course, later, we're going to talk about why open systems

that that rule doesn't apply basically, systems that are open to the flow of energy can resist this tendency towards the disorder. And that's really what the whole story is kind of based on. So through a nested series of such phase

transitions, these are also called these higher order phase transitions are major evolutionary transitions, John Maynard Smith talked about them and meta system transitions by the cyberneticist, Valentin Turchin, basically the same concept with some subtle differences. So the idea is that cosmic self organization is a series of these transitions where functional things come together to make larger functional ones, which come together to even to make even larger ones, and so on. And through this process, adaptive complexity,

or life, so that's a more general way to talk about life becomes better equipped to dominate the cosmos.

Life requires an organization that is increasingly hierarchical and integrated and therefore more resilient and computationally powerful.

So here are just a couple of figures. You see this hierarchy in this structural hierarchy of matter and life. This was adapted from a big physics journal. I can't remember which one it was, but basically, you know, physicists are recognizing this hierarchy, even when it extends to life. So Jeffrey West's book, scale comes to mind. And so we see the hierarchy of science kind of naturally mirror this hierarchy of matter and life. And the reductionist picture would say that these things at the top here are kind of insignificant parts of the story, transient. But we will see that this, you know, the more we go up, the more influence on cosmic evolution, the phenomena have. So Thomas Huxley's question about man's place in

the cosmos is basically the emergence of paradigm, the paradigm of emergence, I'm going to discuss and actually, this is kind of maybe like in a specific or a new paradigm of emergence, because it includes all this stuff about thermodynamics and information theory, says that emergence, the emergence and evolution of life, mind, societies and technology are all part of one thermodynamic process, one evolutionary process, one computational process, unified by the concept of knowledge. So this theory is a theory of knowledge creation. And so knowledge is going to be the unifying theme. And we'll briefly talk about epistemology, I wanted to talk about it more, but it just would have been too long. Cosmic evolution in this paradigm is a universal process of becoming as opposed to being so the universe isn't this static thing. It's actually evolving very much like an organism or a complex adaptive system. So it's not a panpsychic theory in this theory consciousness emerges, that's very important. But you might call it emergent panpsychism. Because as this evolutionary process proceeds, inanimate matter as life spreads,

gets converted into animate matter. So in this picture, humans are neither a cosmic accident, nor the end goal of evolution. So we are instead a step on the evolutionary ladder of becoming. And we are also potentially an essential driver of increasing cosmic complexity. So this big idea is that the universe is undergoing this majestic self organizing process. And at this moment of time, at least in this corner of the universe, we are the stars of the show. This picture is from Eric Cheson's book, *The Rise of Complexity, Cosmic Evolution, The Rise of Complexity*. He's a Harvard astrophysicist. But I just used a Freeman Dyson

quote here because it seemed irrelevant. It's conceivable that life may have a larger role to play than we have yet imagined life may succeed against all odds and molding the universe to its own purpose. So I guess Brian Greene doesn't buy that at all. But if that's true, then as cosmologists have to think about the role of life, there's a lot of people who have said similar things like Seth Lloyd and famously Ray Kerr as well. So just a couple quotes here. So you know, you know, when when people hear these terms progress, there's just the stigma. Immediately, they thought like, oh, someone's trying to sneak in religion, or some intelligent design theory. So you see these trusted names of physics here, David Deutsch has narrowly conceived evolutionary theory considers as mere vehicles for the replication of our genes or means. And it

refuses to address the question of why evolution has tended to create ever greater adaptive complexity, or the role that such complexity plays in the wider scheme of things. So there is clearly implying that adaptive complexity does have this larger role to play. We now see how it is possible for the universe to increase both organization and entropy at the same time, the optimistic and pessimistic arrows of time can coexist the universe can display creative unit erection, you need erection progress, even in the face of the second law, because Paul Davies says that. So that's really kind of a key point. In this is that the second law actually is in a sense driving this increasing progress. So not only can we have unit directional progress in the face of the second law, it seems that this progress requires the second law, because we'll see that the second law is the basically the natural selection pressure for self-organizing systems. So in a sense, the second law is both Shiva the Destroyer and Brahma the Creator. So a goal of this

unifying theory of reality, I've been calling the great consilience, it was actually a name suggested by Marco Lin, who is a is a Fristonian and influenced some of this talk. So basically, the unifying theory of reality illuminates the connection between complexity, cognition, consciousness and cosmic evolution or matter, mind and cosmos. And it's really based on complexity science, it's a lot broader than complexity science. But I guess that name kind of, in a way includes everything. So it's the unification of major sciences of our time, including but not limited to physics, biology, neuroscience, computer science, evolutionary theory and statistics. And basically, it describes complex adaptive systems of all scales in terms of energy and information flows. So it uses statistical thermodynamics, information theory and cybernetics. So we will talk about those a bit coming up. So if you want to see kind of like a a map of what sort of informs this non reductive theory of everything.

Here you go. So I know this is a lot of information. So I'll kind of try to simplify this for everyone.

There are some unifying paradigms that have emerged. So basically, this theory is nothing new. It's basically evolutionary epistemology. After that came universal Darwinism. And later, right now, we're seeing this kind of revolution with like Bayesian inference and these ideas being applied to everything. So being informed by these these paradigms and combining those paradigms are updating them, I should say with information from non equilibrium thermodynamics that we're talking about the thermodynamics of open systems, information theory. So Shannon's theory, some of you will know that information theory is related very closely to non equilibrium thermodynamics and basically Boltzmann statistical entropy is mathematically the same as Shannon entropy or information entropy, which is a measure of uncertainty.

So if I had more time, I would go through thermodynamics and the different types of entropy, but we don't have time for that.

But if anyone has any questions, I can clarify.

Cybernetics was basically information theory applied to biology adaptive systems. It was the first science of adaptive systems, really.

And it's funny, while writing this, I found out because, you know, I hadn't, I had heard the name cybernetics, but it wasn't clear how important it was.

And a talk by Jim Crutchfield, Santa Fe Institute basically revealed that because cybernetics got associated with like the Russian, like during the Cold War, kind of like the Russian agenda, cybernetics was used to design like heat seeking missiles.

Norbert Wiener's theory was almost immediately applied for military uses. And because of that, it stopped being taught in American

colleges. So yeah, it's really important. And it was actually emerging at the same time as information theory, people were already thinking about systems in terms of information flows, but it's more than information theory, but it's a science of feedback.

So dynamical systems theory or complex adaptive systems is kind of like an offshoot of that. All of these inform what I've called universal Bayesianism.

I'm not the only person to call it that. I had searched that term and found Adam Saffron's paper on integrated world modeling theory that used that same term.

But John Campbell has been a researcher who has done a lot to show this equivalence between evolutionary processes and processes of Bayesian inference. We're going to talk about that in a little bit.

John Campbell is a professor of the university. So yeah, if anybody has any questions about that, but really all of these things origin of life research, integrated evolutionary theory.

If you haven't checked out the work of these people, like everybody here is worth reading their work.

So from that we get the unifying theory of reality that is based on what I call the evolutionary epistemology, universal Darwinism, universal Bayesianism framework. That's a long name.

I'm aware of that, but it's just to show that these three theories are basically the same theory.

So universal Darwinism says that the universe is evolving at every scale through both competitive evolution and through self-organization, which is cooperative evolution among agents that have formed a collective

unit.

So it started with Richard Dawkins and his concept of the meme and the selfish gene that memes were these cultural units that were equivalent to genes.

And so we started to see how information was this paradigm that could be extended beyond biology.

And then we had Dan Dennett and Darwin's dangerous idea really extend this to like everything to self-organizing systems to cultural and technological systems.

Evolutionary epistemology came before that. It was based on Karl Popper's philosophy, but it was actually invented by cognitive psychologist, Donald Campbell.

And basically that said that the evolutionary process is a problem solving procedure that creates knowledge.

So the information stored in genomes and in brains and cultural memory.

Basically that all that knowledge accumulates from an evolutionary process.

And it's really helpful to think of it as knowledge, especially when we start to talk about this in the context of thermodynamics and information,

that's the information because basically this knowledge is what allows life adaptive systems to stay far from equilibrium.

They can resist this tendency towards decay. Life can continue to persist if it can acquire the knowledge that it needs to essentially to find free energy that will allow it to sustain itself.

So we'll talk about that in a second more.

So universal Bayesianism is a Bayesian update of the first two paradigms knowledge is encoded.

So the knowledge that evolutionary epistemology can describe, it's actually encoded in biological systems in the form of a world model.

So this model is an internal representation and statistical mapping of the environment, which gets updated through adaptation and adaptive learning.

So we're going to talk about that process.

Really the whole Bayesian brain hypothesis and free energy principle is inspired by cybernetics heavily.

So Ross Ashby's good regulator theorem and the law of requisite variety, which we'll talk about, was kind of the first talk of systems having models and having to model the environment.

So Carl Priden says inference is actually quite close to a theory of everything, including evolution, consciousness and life itself.

We're not going to go into those bigger implications that would be at the end of the book in the last chapter.

But basically, this isn't just a theory about evolution.

Well, it is evolution, but evolution applied way more broadly.

So quantum mechanics has an interpretation that's consistent with this and bigger problems like the fine tuning of the constants and parameters that allow for the emergence of life in the universe.

We can actually probably have the best explanation for the fine tuning problem and the measurement problem of quantum mechanics with this theory.

We won't talk about it, but just so you know, next time, cosmological natural selection by Lee Smolin is the theory that kind of addresses the fine tuning problem from a universal Darwinism perspective, while quantum Darwinism addresses the measurement problem.

So we will save that for next time, but feel free to ask questions about any of those.

So what does this do?

I have this name poetic metanaturalism.

I meant to take that out, but next time we'll get into why I chose that name.

It's actually, it sounds kind of like, you know, like complex or buzzword or jargony, but there's actually very practical reasons for having that name.

So this unifying theory bridges matter with mind.

So universal Darwinism gives us a picture of the open ended complexity growth that this whole idea is based on the idea that evolution keeps producing increasingly complex intelligent forms of life.

Excuse me.

So universal Bayesianism is the link that kind of connects those paradigms with consciousness.

So again, we won't get into that today, but when you understand that knowledge is stored in the form of a world model,

that a system that models itself can experience, then we start to get a better understanding of like what consciousness is.

Consciousness is it's a, it's a mental model.

So it's not a traditional pan psychic theory, even though I know first in and 11, and some of his colleagues have kind of been flirting with pan psychism.

This theory would say that's wrong.

And it's kind of, uh, reduces consciousness to something trivial, um, by saying that it's in everything like, um, a proton is conscious because it's a triad of quarks that have to be in a certain configuration.

So there is some minimal level of integrated information.

Um, that's not enough, uh, to get a conscious experience again, observer.

Um, so Saffron's integrated world modeling theory says that you need a model with spatial, temporal and causal coherence.

I would also argue that it requires self modeling capacity.

Talk about that a little bit at the end if anyone's interested.

Um, but this theory is supposed to bridge mind with cosmic evolution.

So the story starts with the second law.

Um, Daniel, if there's any questions about that, like intro, let me know.

Yeah.

If not, I can just proceed.

Could I ask two questions from the chat?

Sure.

Okay.

The first question is from blue blue wrote.

Do you think that non-living autonomous agents with multiple competing priorities, such as self-driving cars have free will?

If not, what is the defining difference between these systems and humans?

Uh, no, I don't think they have free will.

So, um, adaptive information is a special type of information.

I do think those things, and this is kind of following Sarah Walker's lead, um, who's at Arizona state.

I think she's also SFI faculty now, but basically, um, to have self-driving cars, you first need life.

So anything like a self-driving car, that's like this information processing system.

Uh, if you were to come across, uh,

uh, broken down self-driving car on some planet, and you saw that you would have to infer that there was also life there because life is part of the trajectory to get to that.

But I wouldn't say that it has free will because it's just this input output machine, uh, with a defined set of algorithms and, uh, adaptive systems are more flexible.

Uh, and there, okay.

So in the next, next time, we'll talk about actually the emergence of agency, uh, which occurs with the origin of life or abiogenesis.

And you basically, um, because this self-organized system that would be kind of like the protocell, um, is evolving through a mechanism that a Darwinian dynamic for self-organization that we'll explain in a bit, it builds up information.

And at some point there's a phase transition where the information that's getting built up in the system, uh, basically, uh, gains what Sarah Walker and Paul Davies call informational control.

So there's a point where information actually starts calling the shots.

And you can see the difference between, uh, systems that where there is informational control.

Uh, so living systems and inanimate systems, because for example, um, inanimate systems, uh, their movement can be predicted with Newton's laws.

So anyone who's taken first year physics, uh, probably had to do an exercise where, uh, you're trying to predict the movement of some macroscopic system, like a ball that gets kicked or pushed by a gust of wind.

And what you do is you, um, you have to draw out all the forces acting on the system.

And then you can understand where the system's going to go with life.

You don't need an external force for it to start, for example, climbing uphill.

Um, that's internally generated.

Of course, it's not magic.

Uh, the adaptive system, the agent, uh, has stored energy that it's extracted from the environment.

So for example, humans that have eaten food and there's a metabolic process that's driving this, but, uh, still, yeah, you, with living systems that where information has causal power, uh, basically the trajectory of the system can't be predicted by these fundamental laws of physics.

Um, Philip Ball, uh, journalist that has been talking about this, um, his recent article, which caused a lot of controversy between like the reductionists, like talking about free will and agency.

Um, and there's a lot of confusion around that.

So people like Jerry Coyne were like trying to shoot it down and yeah, hopefully I can talk about that.

Um, I guess we'll talk about that next time because there's a lot of, uh, confusion about what we're saying when we say something has agency or free will.

But his example is if you throw, throw a ball and a bird off the tower and you can predict what one is going to do.

You can't predict what the other is going to do.

Um, interestingly though, I would say that the bird's not completely unpredictable.

You can predict that it's not going to hit the ground.

It's not going to splat on the ground.

It's going to fly away.

Uh, you have to understand the bird in terms of its goals.

So you have to understand, you know, the bird and its environment and the information that got built up through evolution.

And then we can start to predict the bird, uh, its statistical behavior.

So inanimate systems, you can pretty much precisely predict the movement unless they're chaotic systems, which is another story.

But with life, um, yeah, it's not predictable in that same way with like low level laws of physics.

But, uh, I'm arguing that actually life is more predictable than we thought.

But you need things like, uh, non-equilibrium statistical mechanics and for example, uh, free energy principle.

Um, so you can say that, you know, an organism will try to minimize free energy or, uh, minimize the difference between, uh, its models prediction and reality.

So it'll try to minimize prediction error.

And that's a way that we can start to describe, uh, behaviors statistically.

So, um, yeah, was there another one?

Yep.

Can I ask the second question?

Yeah, I know I didn't completely answer that, but, uh, that will require like the, all of the points that I would get into in the second talk, but it's a great question.

Yeah.

Yeah.

I would have written in a larger margin if I had more time.

The second question is from Joseph Clark, who wrote, uh, first complimented your talk and let you know that you were an interesting guy.

So then the question was how far do you think the universe can organize itself?

And what does that look like?

Okay, great.

That's a perfect question to lead into the next part.

And hopefully by the end of this, uh, we will have some sort of answer for that, which is good because I, when

I started making the beginning of the presentation, I was going to get to consciousness and free will.

And I ran out of time.

So there's been a lot of talk about things that I won't get to, but that we will get to.

Um, so, uh, the story starts with the second law.

Um, so the second law of thermodynamics as popularly, popularly understood says that, uh, systems naturally become increasingly disordered.

Um, that's really just because of the large scale effects of chance.

So there are many ways to be disordered and there are relatively fewer ways to be patterned.

Um, so a system that's this collective of particles.

So this evolving ensemble of particles, uh, will naturally tend to move towards a configuration that's completely spread out and mixed up.

There are no, uh, energy gradients.

So nowhere it can be extracted.

That's a state of equilibrium.

So we are going to use this word thermodynamic equilibrium, uh, to mean death in disorder.

So life wants to resist the tendency towards equilibrium.

It wants to stay ordered.

Uh, what's interesting about this is that it applies to all.

This is kind of like where the free energy principle.

And for instance, Bayesian brain hypothesis, it, it, it frames everything this way.

So I thought it was a good way to kind of frame it here is that you start with the second law in any conceivable system, any ordered system for it to continue to persist has to resist this, uh, tendency.

So, um, basically, uh, Schrodinger.

Uh, Schrodinger.

The guy everybody knows from quantum mechanics and the, and the cat thought experiment.

And he wrote a book called what is life in the forties.

Uh, basically kind of solved this paradox.

Like if there's the second law and things become more disordered than what's with life and what's with the biosphere and all of this complexity that we see around us.

And he explained, actually Boltzmann explained this before him.

Um, but Boltzmann said a lot to contradict this as well.

Um, or just didn't follow this to its implications.

But, uh, Schrodinger explained that open systems, uh, by feeding off, uh, free energy in the environment.

So free energy is just energy that, uh, you can extract work from.

So it's ordered energy as opposed to energy that's been dissipated in the form of heat.

You can't extract energy, uh, uh, useful energy from that energy.

It's still there.

It's just spread out and it's, it's, you can't harness it.

Collecting it would cost more than you'd get out of it.

So, um, Schrodinger explained that, uh, if a system can get free energy, which he called negative entropy or energy, uh, negentropies was called by, uh, someone later.

Um, so to understand how ordered systems stay ordered, uh, you basically need to understand that they have to, uh, constantly be extracting free energy from the environment.

So, um, basically, uh, the second law is not violated by life because, um, in this effort to stay far from equilibrium life, it keeps extracting that ordered free energy and keeps dissipating it as heat.

So, uh, so it's basically exporting entropy into the environment.

So the order, um, that is maintained, uh, is paid for by the dissipated energy, which come becomes thermal entropy.

So there are multiple definitions of entropy.

So you have thermal entropy, which is heat entropy and then statistical entropy, uh, which explains thermal entropy, but it has a broader domain of application.

So you can talk about shuffling a deck of cards and you can talk about like the statistical entropy increasing, uh, if the deck is ordered and it starts becoming mixed up through the shuffling.

But the deck of cards, the thermal entropy, uh, isn't changing much at all.

So there's different types of entropy.

There's also information entropy, which has to do with the, um, uh, the number of messages that can be sent across the channel instead of, um, the number of states that a system can be in.

The number of microstates without changing the macro state macro state or, uh, uh, the kind of, um, collective properties of the system.

And there's a really neat relationship between statistical entropy and information entropy, which I won't talk about here.

It was really fleshed out by, uh, E.T.

Janes.

Uh, but yeah, so information and entropy like go hand in hand.

Um, just to give a little more context before we get into the details, uh, David Deutsch in this great TED talk, um, called after billions of years of monotony, the universe is waking up.

Says if one can speak of a cosmic war, it's a war between monotony and novelty between stasis and creativity.

And in this war, our side is not destined to lose.

If we choose to apply our unique capacity to create explanatory knowledge, we could win.

So that's very different than Brian Greene's story.

And it kind of gets at the second question that was asked there.

How, how far can this complexity increase go?

And so this kind of foreshadows that maybe it doesn't have a limit.

And that seems, you know, most people think that would go against the second law of thermodynamics.

We're going to see why that's not necessarily true.

Um, but you can also see this cosmic war as this fight between order versus disorder, life versus entropy and knowledge versus uncertainty or ignorance.

Um, but is life's battle against disorder, uh, good versus evil or yin and yang?

So I'm going to kind of argue that these two things kind of have this, uh, you can describe their relationship as causally dependent.

And it's kind of like this dialectic between life and, uh, disorder.

Um, and it is through this sort of, uh, interaction that, uh, creates progress.

So we have some Eastern ideas mentioned again.

Um, but I, when I was writing the book, a friend that practices Buddhism called me and told me about this interdependent co-arising idea, which I thought was really cool.

So we're going to come up with this, what we call Popper's principle, which is, uh, problems create progress.

We're going to see that this problem of entropy, that this tendency towards decay is what actually, uh, forces systems to evolve.

Basically it forces them to search what's called the configuration space for the phase space for the state space to find configurations that are good at extracting energy.

And that allows it to stay far from equilibrium.

So we're going to have some figures soon.

So this stuff that's kind of abstract becomes a little bit more concrete.

So, just out of respect to the earlier request, how about you check the time and feel then how fast you'd like to continue, however much you want to present.

And then if you want to take any questions, but first just check the time and let just,

I, I, how I didn't, uh, I didn't, uh, notice when we started.

So about how long have I been going?

I would say five, zero minutes.

Really?

Okay.

How long can we go?

We can go another, uh, 99 minutes.

Okay.

Um, I'll try to go through it a little bit faster.

Um, but yeah, so, um, but if we use all the time, I told you yesterday that I was feeling, uh, oh, I had a sore throat and, uh, the sore throat's better today.

I'm hopped up on cold medicine.

I'm sure I'm going to, I'm going to pay for it tomorrow.

Um, but, um, yeah, if we use all the time with questions, that would be good.

Uh, so.

Um, so, uh, in our everyday experiment experience, things don't naturally become ordered unless someone intervenes.

Rooms get messier buildings, the road.

Uh, so we already talked about this Schrodinger's solution.

So we can think of life as a game.

Uh, so life has always been playing a game against the second law of thermodynamics.

There is another narrative that I won't go into, but life is also an energy flow channel.

Life by existing and computing and trying to stay far from equilibrium is also increasing entropy.

Uh, actually the universe of life will increase entropy at a faster rate than a universe without life.

Um, which is pretty interesting.

So it's again, not necessarily that life is like battling against entropy and in many ways it increases entropy, but there's also this story, I guess what it's trying to battle is the tendency towards disorder.

So it depends on what type of entropy you're talking about.

Um, so the challenge in the game of life, which applies to you as well as other simpler adaptive systems is to resist that tendency.

So this gives life a goal or a teleology.

So teleology was associated with kind of this mystical thing, but basically we'll, we're seeing that teleology is just, uh, agency.

It's just a result goal oriented behavior as a result of the information that's been accumulated over evolution.

So teleology, uh, the book, uh, tries to naturalize teleology and just say that there are these goal oriented systems that behave different, uh, because they are acquiring information through Darwinian evolution.

Um, so life has to attract free energy.

That's how it stays in the game of life.

And to extract free energy, it's, it's not a simple task.

So the computational task of extracting free energy, uh, requires that the, uh, system acquire information about its environment.

Otherwise it can't find energy.

Uh, it also requires that the system model its environment.

So the information is creating this predictive model, this generative model.

And it is, um, that model that allows it to continuously find free energy.

Um, so we will explain that life strategy is it searches through a configuration space for solutions to the problem of survival through trial and error search, uh, variation and selection.

So we, we know about evolution.

There's this genetic variation in natural selection.

And, uh, we're gonna argue that that process, that mechanism is much broader.

The evolutionary mechanism is actually the mechanism of the scientific method and of adaptive learning.

Um, and that mechanism is also a process that accumulates evidence-based knowledge.

So it's also a process of Bayesian inference.

We're gonna explain why.

So here we have those different mechanisms with different names.

Um, so it's not included here, but, uh, so if life is a game, there are levels to this game.

Uh, and the levels we're gonna see are these, uh, evolutionary transitions.

Every time life graduates to this, uh, more, uh, complex, uh, hierarchical system, like when single-celled organisms come together to form a multicellular organism and multicellular organisms come together to form

communities like societies, uh, that you can, uh, view these evolutionary transitions as, as life graduating to a new level.

So, um, we're gonna talk about how adaptive information gets built up through evolution and, uh, we're calling that knowledge.

It's nice to have that word knowledge because this process of an adaptive system acquiring this information is actually reducing its uncertainty about the environment.

So it's, it's reducing its ignorance about all the ways the surrounding world could potentially surprise it.

Uh, so knowledge is a nice word.

Some philosophers, like old school philosophers might have issue with that.

Um, because in some teachings, like knowledge is something that like conscious beings with like awareness have, but we're calling any adaptive information knowledge.

And it's that more of a technical definition because knowledge is that information, which reduces, uh, uncertainty.

And you could actually quantify this process of evolution.

Terence Deacon has talked a lot about how you can do that with, uh, information measures.

Um, so just to explain this, why is knowledge something that matters?

Like, does it have causal power or is it all just this, this billiard ball universe where it's like these particles bumping into each other?

And you can basically use, uh, fundamental laws of physics to describe like the trajectory of all the particles in the universe.

This is saying, no, that, uh, the information that gets built up through evolution has causal power, such that these systems start to behave in ways that are different than what would be predicted from, uh, something like Newton's laws.

So we also have this process of anti accretion that Sarah Walker likes to talk about.

Uh, but anti, so anti accretion is, uh, basically we're familiar with, uh, planets pulling in matter.

So bodies like asteroids, because, uh, gravity pulls matter in naturally, but anti accretion is when a planet is ejecting matter.

And, uh, in many cases, what, well, so the, the only time we really see anti accretion, uh, is when you have a planet with life on it.

So, um, right now, uh, bits of biosphere.

So humans and space rockets are being sent to other celestial bodies.

So that's a form of anti accretion, uh, sending up satellites.

You won't see a satellite get sent out.

If you see that in the universe, uh, like you're looking at some other solar system and you see a body, uh, sent out and put into orbit from a planet, that's an indication that life is there.

That's a bio signature.

Um, so she says that anti accretion requires comprehenders specifically the existence of physical systems with knowledge of Newton's laws.

Um, so what does it all mean?

Um, it means we need knowledge, uh, to, uh, keep us out of thermodynamic equilibrium.

Um, so how do we acquire knowledge?

Uh, science is the most, uh, is the most salient example of the causal power of knowledge.

So science, uh, has, uh, eradicated lethal diseases, built a global communication network called the internet and created weapons of apocalyptic power.

It's not always good, this power of knowledge, but, um, uh, we see that there's something about the scientific method, which is efficient at, uh, accumulating knowledge.

So we're going to talk about why, and that's where we get into Karl Popper and evolutionary epistemology.

Um, so we know knowledge is important.

Karl Popper says we're always faced with practical problems and out of these grow sometimes theoretical problems for we try to solve some of our problems by proposing theories.

Karl Popper stresses that all of science starts with a problem, either theoretical or practical.

Um, and it is that problem that forces us to seek out a solution.

So we can already see why this principle that I've called Popper's principle, uh, uh, says that problems create progress.

Karl Popper says we're going to be in progress.

So some of these theories that we make to try to solve our problems will be wrong.

They'll be errors.

Um, but the idea with science is that those errors get filtered out by a process of testing your theories and a peer review process, which gives criticism.

So, uh, knowledge is really, uh, for knowledge, for us to know that some piece of information is knowledge.

It really needs to be tested.

And, uh, we call that evidence-based knowledge or evidence-based information.

So it's really the only true kind of knowledge because you don't know if something's true until you test it.

For example, um, the world doesn't look round from a naive perspective.

It looks flat.

If it looks flat.

So for example, um, someone who had the first idea, you know, and I'm sure it was before anybody famous.

We knew someone had this idea and probably people blew it off.

But, um, uh, some things that are true about reality are not immediate immediately apparent from like sensory observation from since data.

So we really need to test our theories if we are going to be sure about them.

Um, so what Popper found out was the scientific method is an algorithm.

He called that algorithm conjecture and refutation.

And, uh, basically this is the method of hypothesis testing.

So you have a problem you believe can be solved.

You make an informed guess or conjecture to see if your theory can be refuted, uh, prove false by testing its predictions.

So, uh, people have pointed out like John Campbell, uh, Karl Friston, that, uh, science is a process of inference.

So he recognizes statistical patterns and trends in nature that we can use to make increasingly accurate predictions.

Um, it would not be an accurate thing to say that the function or purpose of science is to generate predictive knowledge.

Um, and we can characterize that as process of inference, which basically means that scientists draw logical conclusions about the way the world works,

uh, based on evidence based information acquired in the past, but with never present awareness that the information can and will lead to,

um, to new conclusions and deeper understanding.

So when we recognize this, we also have to recognize that all of our models will have uncertainty.

Um, so we should never think that our model's right.

Every model is going to be proved wrong in some way, but through the scientific process,

uh, the idea is that we get closer and closer to truth.

So we can never reach perfect truth, but in this model, you can actually get closer to it.

So in some senses, it's different than like postmodernist philosophy that says there is no objective truth.

There is an objective truth.

Um, uh, but we can never know it, but we can get closer to it.

Uh, so once Popper understood that knowledge, the knowledge generating mechanism behind science success was conjecture refutation.

He realized that human learning, which begins at birth and continues to death uses the same problem solving algorithm.

Although the developmental psychology literature called the method trial and error.

So because life is constantly presenting us with new challenges, such as the need to get from one place to another,

a problem that inspires babies to learn to walk.

We must constantly try out new behavioral solutions.

We can think of these actions as guesses about how to survive, or if you prefer experiments innovating equilibrium or predictions for persistence, and they will often fail.

Um, so this is Dan Dennett's idea of life exploring this design space.

We're going to talk about how it does so through evolution, but we're still talking about science here and adaptive learning on an abstract level.

We can imagine a certain practical problem as a challenge one with the solution that exists somewhere out there in the space of possibilities, just waiting to be found by someone sufficiently motivated and clever.

It may take some time, but if the possibility space, the space of possible solutions is continuously explored in an efficient way, eventually a solution will be discovered.

So, excuse me, trial and error learning explores the solution space.

So just to give you a clear example of how adaptive learning is kind of this form of hypothesis testing.

So whenever we have a problem in life, and we don't know the correct solution to a problem in advance, we naturally begin with those potential solutions that are closer to our starting point in the solution space for sampling.

So for example, if a baby reaches for its bottle and barely misses, it'll adjust its behavior slightly, which minimizes the chances of making an error.

So our instincts may nudge in the general direction of a solution, but the instinct alone is not enough.

We have to try out what you could call a behavioral conjecture, and when it fails, we try something new.

So that's kind of like a new theory.

And it's essentially the old one, but with a little twist.

And if that behavioral solution is remembered, then it gets stored for future use, ready to be repeated, but also adapted if necessary.

But in humans, so the baby example reaching for the bottle, let's say it misses the bottle, it won't get a reward signal.

It won't get a little surge of dopamine.

So if it corrects and it gets closer, it gets a little reward signal.

This is reinforcement learning.

But basically this adaptive learning process is a process of making theories and eliminating those theories that are errors.

So Popper saw that scientific knowledge and the evolutionary process were also connected.

So not only is science adaptive learning, both of those things are extensions of the evolutionary process.

So he says, in science, theories are highly competitive.

We discuss them critically, we test them, and we eliminate those theories which we judge to be less good in solving the problems which we wish to solve.

So only the best theories, those which are most fit, survive in the struggle.

This is the way science grows.

So that's kind of nice, like showing you really how this is an evolutionary process.

Not only is science an evolutionary process, the converse is true.

Evolution is a scientific process.

So it is clear, this is another quote, it is clear that this view of the progress of science is very similar to Darwin's view of natural selection by the way of the elimination of the unfit.

So it's clear that this is a theory of the evolution of life, the errors in the attempts of adaptation, which is a trial and error process.

Analogously, science works by trial, theory making, and by the elimination of errors.

So I might pause in a second just to see if everybody understands this, but I have a few more slides.

So evolutionary algorithms also use this mechanism, but it's called generate and test.

So I didn't mention this, the mechanism of evolution is variation and selection.

So we have these algorithms that are equivalent conjecture and error mutation, trial and error, variation and selection.

We also have generate and test.

So machine learning evolutionary algorithms have this method generate and test.

Possible solutions are generated into an actual solution is found.

And these solutions accumulate in some memory store while the errors are filtered out and forgotten.

In science solutions are theories or models that accurately predict some natural phenomenon.

And the successful ones accumulate in peer reviewed journals and textbooks.

So we see that these processes are equivalent or actually going to see that it's this Bayesian process.

That adaptation is also a process of the model of the organism becoming optimized, decreasing its prediction error

so that it can more efficiently extract free energy from an itch to stay ordered.

So we get this principle that all evolutionary processes are learning processes and all learning processes are evolutionary processes.

Conrad Lorenz, a Nobel Prize winning zoologist, was also a pioneer of evolutionary epistemology.

And he made the statement life is a cognitive process.

So if this is true and there's this functional equivalence between the mechanisms driving evolution, learning and science,

then that implies that adaptation or the genetic information that corresponds to those physical adaptations are equivalent, is equivalent to scientific knowledge.

They're actually the same thing. So to kind of make this relationship clear.

Biological adaptation represents knowledge of the environment and the knowledge we acquire through learning and science reflects adaptation to the environment.

So we see that there's no meaningful distinction between adaptive information and scientific knowledge, both allow life to predict an uncertain world, control matter,

constrain chaos, constrain chaos, and construct order from disorder.

So the very same property that is responsible for organic matter leaving the planet in spaceships.

So this anti-accretion is the same property that allows organisms to climb uphill, seemingly defying the force of gravity.

Of course, the laws of physics aren't violated in any way, but living systems are not constrained by the laws of physics the way inanimate systems are.

So some examples, a dolphin streamlined design, which is a product of the information stored in its genome, contains the knowledge of hydrodynamics.

An eagle's wing design contains the knowledge of aerodynamics.

An eagle's wing design contains the knowledge of aerodynamics.

Not only can we be certain that engineers see knowledge in these functional structures, there should be little doubt that they inspired our machines.

And without that information, I don't think we would have come to those inventions, at least not any time soon.

Camouflage on an organism represents knowledge of the environment.

So kind of bridging this idea of evolutionary epistemology that evolution builds up knowledge and that science is an extension of the evolutionary process.

So making a bridge between that and Bayesian inference requires this relationship between adaptation and statistical correlation.

So physicist Carlo Rovelli, who has gotten really interested in explaining how this information that underlies agency, that behavior that we see with living systems, gets built up through evolution.

He wrote a great essay.

He wrote a great essay.

He won the Foundational Questions Institute essay contest a few years ago.

But this work was based on the work of David Wolpert and Artemy Kolchinsky and basically shows how this adaptation builds up information in the system.

That information is really statistical correlation between the organism and its environment.

So here's the simplest example.

It's a bacterium performing chemotaxis.

So a bacteria, so chemotaxis is basically a bacterium will swim in the direction of food and swim away from toxins.

So basically it's detecting this chemical gradient that follows.

But so we can see in this example, and this isn't the case for all species, but you can imagine that a bacterium that swims to the left when nutrients are on the left,

prosper compared to a bacterium that swims at random.

So with this behavior where it's swimming towards food, that is the product of a chance mutation.

So in some sense, it's an accident.

But when it discovers that solution, that organism is more likely to reproduce.

So that solution gets retained in the population, it gets sort of frozen, hardwired in.

And so the process of adaptation is a process that builds up a statistical correlation between the system, the agent and its environment.

And so that shared correlation represents mutual information.

So mutual information is information that's shared between the environment and the organism.

And just to explain that a little bit more, this correlation, basically, as this correlation gets built up, it's creating predictive knowledge.

So John Maynard Smith cites Fred Dretskey, who did kind of the pioneering work applying Shannon's information theory to try to quantify evolution.

So Dretskey argues that if some variable A is correlated with the second variable B, then we can say that B carries information about A.

For example, if the occurrence of rain is correlated with a particular cloud, then the type of cloud tells us whether it would rain.

So that's from a great essay called the concept of information biology.

So we can see that you have this little rhyming equation here.

adaptation equals statistical correlation equals mutual information.

And because that shared information allows the organism to predict its environment better, this process is also a process of model optimization.

So the predictive model of the organism as it adapts starts to become more accurate.

And so this whole process can be seen as a free energy minimization process.

Of course, that's information theoretic free energy.

It's the free energy of, for instance, principle, we're not talking about thermodynamic free energy.

Anyone wants to ask a question or gets confused about that.

Let me know.

But it's basically those things are related.

It's something that's not mentioned enough.

Minimizing information theoretic free energy, which is just minimizing the prediction error of your model is what allows the organism to minimize environmental free energy.

So by acquiring information, reducing the information theoretic free energy allows the organism to extract energy and reduce thermodynamic free energy.

It minimizes the free energy in the environment and converts it into entropy.

So we kind of see how this is getting into universal Bayesianism as it becomes more correlated.

It gains information about its niche and therefore becomes a better predictor of that niche.

So that's what adaptation is.

It's the genome of the species accumulating information that reduces the average organism's uncertainty regarding the environment.

So the genome is a knowledge repository of solutions to environmental challenges that allows an organism with no brain and no mind to anticipate events in the environment.

So in this theory, as I mentioned before, consciousness comes later.

It comes with self modeling capacity and a bacterium would be kind of a zombie agent.

It would have agency and it would move with purpose.

But that is not enough for mind or at least mind defined as consciousness.

If you want to say that mind is just information processing, then, yes, there is mind in these systems.

But that's very different than saying that the system has a subjective experience and a first person perspective that will come in the next talk.

But that it's it's harder.

So that information sort of memory is integrated into a statistical or predictive model that determines the causal control systems behavioral repertoire or the full range of behaviors an organism is capable of.

And it is this mapping of sensory inputs to motor outputs that makes the organism move in ways that we normally associate with mind or consciousness.

I'm Karl Friston in a great Eon article says all biological processes can be construed as performing some form of inference.

I think I already read that.

But the free energy principle for those aren't familiar.

We've already explained it, but the Bayesian brain hypothesis and the free energy principle are the same thing.

The difference is that the free energy principle applies to all other adaptive systems where the other one is just focusing on brains.

But first and realize that this is a model and a process that applies to any system that's trying to resist the tendency towards disorder.

So any system to stay to continue persisting longer than we would expect naturally, it must engage in Bayesian like learning to deal with environmental uncertainty.

How much time do I have left roughly?

Still more than enough.

I'm definitely more than half through I'd say I'm two thirds of the way through.

So, John Campbell has this great paper called Universal Darwinism as a process of Bayesian inference that I kept coming across as I was trying to understand non equilibrium thermodynamics and information theory.

I should say actually, I should have done the books, showed you each book, but there's a great book called The Origin and Nature of Life on Earth by Santa Fe Institute's Eric Smith and Harold Morowitz.

Morowitz was a professor of mine.

Eric is a friend that we would have these foundations of the mind guild.

It was like a consciousness club that would meet at the Krasnow Institute where I got my PhD.

And so their book, The Origin of Life on the Origin of Life in Chapter 8 actually gets into this information theory and this Bayesian inference and describes the origin of life and the evolution of life as a process of Bayesian model selection.

And so that was kind of what connected non equilibrium statistical mechanics to these to this Bayesian brain hypothesis.

And I realized that, you know, these theories were kind of converging on the same ideas.

So John Campbell wrote this paper in 2016.

He says Bayesian inference is an algorithm for the accumulation of evidence based knowledge.

This algorithm is now seem to operate over a wide range of evolutionary processes, including natural selection, the evolution of mental models and cultural evolutionary processes, notably including science itself.

So when we say that evolution is a process of Bayesian model selection in the book I mentioned with Eric and Morowitz actually characterized evolution as an inference engine.

So we're basically saying that when organisms that aren't fit get weeded out.

That's basically models that aren't predictive getting eliminated.

So the organisms that persist are the models that were able to predict the environment best.

So inference and prediction are kind of the same thing since the word inference is a little obscure, I guess.

So just to make this point a little more clear, since nature is complex and intrinsically unpredictable, the model of an organism is at some point guaranteed to fail when reality surprises it with the unexpected.

So that's what the free energy principle is about.

Minimizing surprise is the same thing as minimizing prediction error.

So an organism gets eaten by a predator, baby fails to reach his bottle, scientific theory fails to explain new data.

Basically, when any of these things happens, it means that the model used to predict reality is not completely accurate, it contains a certain amount of prediction error and needs to be corrected.

So what's the solution? Try something new, the old model worked well enough up to that point so don't ditch it, just vary it a bit and see if it performs better.

better. If it does not reduce surprise, eliminate that variant and try again. If prediction error

is reduced, replace the old model with the new one so that it becomes the reigning theory and

the design template for new variants to be based upon. When a model has been updated in this way

due to natural selection, adaptive learning, or experimental testing, we can say that knowledge has been acquired and environmental uncertainty reduced. So here we have an image. I should say that Infinity Maps, a German software mapping company, helped me create these images. Oliver Wan, I have to give a big thanks to him. This image has a little bit more going on than I would want right now, but you could see that basically this represents two different mechanisms for learning and adaptation. So this one on the top that we're going to talk about first is the normal competitive evolution, the sort of Darwinian survival of the fittest, while this mechanism over here is the mechanism of self-organization. So let's first talk about the normal type of evolution we're familiar with and show that how this process of evolution that works through variation and selection actually generates predictive information and creates a model which is which encodes knowledge of the biologically relevant regularities in the environment. So first we have this, this is a bacterium over here and so this is the simplest model. You have an organism that's an embodied theory or a best guess about how to stay far from equilibrium. So evolutionary epistemology sees organisms themselves as embodied theories. So evolution is basically testing these different models. So it basically reproduces and we know that because there are going to be errors in that process, the genome will get changed slightly and you will get a generation of progeny that all are slightly different. They have slightly different configurations. Some of these configurations will be better at extracting energy and predicting the environment than others and the ones that aren't good at predicting the environment, they get filtered out by natural selection. So here we see design four is the winning design because it's the best predictor and it's the best predictor because it's the best extractor of energy. And so that design that that organism gets to reproduce and then you have another generation of designs that are slightly different and this process is an iterative process just like science and hypothesis testing. So over generations you will start to get designs that are better adapted to the environment which we've explained means that they will be more correlated with the environment and being more correlated means being a better predictor.

So the knowledge that's left over after this process is the adaptive information stored in genetic material that reduces environmental uncertainty and you see that this process will ultimately lead to the evolution towards the the best possible design.

So it's it's an optimization. A lot of people don't think that evolution is optimizing anything. There's been this, you know, kind of philosophy that evolution does just good enough. But we actually see that in terms of the computation that it does to stay far from equilibrium, that it's actually optimized for being almost maximally efficient with its use of energy. It approaches something close to what's called the land hour limit, which David Wolpert at SFI is written about. But life is much more efficient with computation in terms of energy supply needed compared to all of our technology. So the human brain can do solve problems that even our best AI systems can't solve and it does so running on the equivalent of one household light bulb.

So we'll come back to this self organization process. Oh, well, actually, we'll get into that right now. So this is the normal process that we think about when we think about evolution generating knowledge predictive knowledge.

And so that's called phylogenetic learning. Phylogenetic refers to these like generations. So learning is generational, the organisms don't really learn, because they don't have brains, they're not able to update their model in real time. So learning is at the level of the population.

And for learning to occur organisms have to die before brains. Basically, you have to have this competition for there to be a learning process. However, there's also self organization when these organisms don't compete, but actually find a configuration find it discover collective configuration through their somewhat random interactions with each other.

And when they find this collective configuration through trial and error.

Basically, the configurations that are better at attracting energy will be the ones that are retained. So Donald Campbell, father of evolutionary epistemology, explained this process, this evolutionary process, by which self organization works as blind

variation and selective retention, or we can think of it as selection for persistence or stability. So fitness kind of

there's a Dawkins quote, I think I come out where Richard Dawkins said Darwin's survival of the fittest is really

just a more specific instance of a more general law of survival of the stable. So when we think about being fit, at the beginning, we're talking about

just being stable, being able to find energy, if you can't do that, it doesn't matter whether you have agents that you're competing with or not. Because if you can't get energy, it doesn't matter if there are other people competing for it, that you won't survive. So you don't need replication with variation to evolve. That's what this is this model is saying.

And Stuart Kaufman from the Santa Fe Institute has been talking about the importance of this marriage between Darwinian evolution and self organization. But basically, because there's not this replication process.

And this is how evolution occurs. So you have a system that's a collection of components, parts, these parts can be molecules, or they could be agents. And when these molecules are pushed far from equilibrium by flow of energy, of course, humans are eating food and metabolizing energy. So they're being powered by energy flows as well.

The system blindly explores various configurations in the space of possible designs via trial and error. So the system will move through a series of configurations. And those configurations that allow the system to extract sufficient energy to stay out of equilibrium will get retained, while the configurations that don't allow for energy extraction will get filtered out either the system will collapse and have to try to

new configuration. Or it will just revert to the previous stable configuration. And it will continue trying to explore that configuration space. I should have said, like, here's a good example, like, here's a depiction of this configuration space, we're talking about life exploring the design space. So basically, you have a designer configuration, and then

you get these different, you get these different, this exploration of different configurations that are very close to the configuration you started out with.

So you know, I was thinking about giving a talk on the origin of life, but I couldn't get into all this cool Bayesian stuff. That's what the first part of the book is about. What's really interesting is that refining that self

organization, as mentioned, is a Darwinian process. So there's actually a name for this dissipative adaptation, and it shows that this evolution towards these more stable configurations dissipate more energy. When this happens at the level of organisms that we're talking about, it shouldn't be surprising that it shouldn't be surprising that it dissipates more energy, because basically, as the organism evolves to be better at extracting energy, all that energy it's extracting to maintain far from equilibrium is being converted into thermal entropy. So all self organizing processes are really dissipative adaptation processes at different levels. So this process, philosophers would say is multiply multiply realizable. So this process can happen at different scales. And that's exactly what we see like, right here, the same process. So you would have like society, like exploring different governance systems. And it doesn't know which government system is going to work the best beforehand, it tries it, if it doesn't work well, like the fall of Rome, it will collapse, and then it will reassemble.

Reassemble. So just to wrap up the origin of life. So we have this process of dissipative adaptation. And if it's being pushed by the flow of energy, and it's doing this process here, where it's finding increasingly stable configurations, collective configurations, then basically, basically, basically, what will emerge is an auto catalytic set, we'll define that later. But basically, it's an, I'm losing a lot of terms from philosophy, I was going to say, like, it's the beginning of an auto poetic agent. So like a self maintaining agent, and its simplest form is a chemical system, and its simplest form is a chemical system that produces outputs, which are then fed back into the system, so that the system can self amplify, it's basically creating all of the molecules that it needs to grow. And it's thought that this process, you know, occurred, the auto catalytic set is kind of the precursor for the first cell. So it's kind of like a proto cell. And it was thought to occur around hydrothermal vents, which are underwater volcanoes, there's all this energy flowing up from the the hot core, running through geochemical energy.

And there's these rocky pores in the rocks around hydrothermal vents, which provides something like a membrane, like a lipid membrane to hold the components in a Markov blanket. So the pores in the rocks were the Markov blanket for the auto catalytic set. And Eric Smith and Marwitz showed that basically, basically, feeding off inorganic inputs, like carbon molecules, like not not anything, not organic chemistry, where it's like life, but just like, simple, like carbon, hydrogen and oxygen feeding off those inputs. Basically, the metabolic process, it's called the reverse Krebs cycle. So that process allowed the first auto catalytic set to continuously self amplify through this through this process. So if this process is allowed to continue, further self organization may eventually lead to the emergence of an organic computer that can process integrate and store information about the world. So what dissipative adaptation is doing is, as it's doing this blind variation and selective retention process, and it's finding more stable configurations, those more stable configurations, those more stable configurations are also predictive of the environment. So this dissipative adaptation process is this auto catalytic set, modeling the surrounding energy landscape. So I'm arguing that this is where modeling begins, it even begins before what we call life like rep self replicating life, and it's dissipative adaptation, which is a process of modeling the the energy in the environment.

Dr. David P. So this isn't just one continuous process. There are these leaps that are phase transitions. Basically, we've heard of phase transitions and physics and chemistry courses where you have like, Dr. David P. Water being heated up and then transitioning into a gas, or the reverse when a gas becomes frozen. And then you have a gas

Dr. David P. Water being heated up and then you have a gas, or the reverse when a gas becomes frozen. You get a state of order that emerges. These transitions towards these configurations that are good at extracting energy. Those have also been characterized as phase transitions. So Eric Smith, who I mentioned before, and Harold Morowitz, they have this phase transition theory of life, which basically explains how information

Dr. David P. gains control in these systems. And Sarah Walker and Paul Davies work on this basically showed that through this process, the blind variation and selective retention, information gets built up in this auto catalytic set. And at some point, there's what they call algorithmic takeover, or informational control, and the system becomes an agent that can start to steer and control itself.

Dr. David P. In ways that wouldn't be predicted by physics. So living system is very much like we didn't talk about dissipative structures, but like tornadoes, whirlpools, those things emerge to dissipate an energy gradient.

Dr. David P. Living things because of the process of metabolism, we are this cycling system, this thermodynamic energetic loop.

Dr. David P. But what was I going to say about that? So yeah, we are like the storm, but we do have control over the storm. We're a storm with control. So it doesn't really make sense going back to Sam Harris's quote that you are the storm, you do not control it. It's literally the information that gets built up through evolution that allows us being this dissipative

Dr. David P. So the last part of the storm to start controlling its own behavior.

Dr. David P. So the last part is just an argument for that second question, why should complexity increase?

Dr. David P. We know that this evolutionary process where you build up this model.

Dr. David P. So we would think that this after these generations of evolution, this organism is going to be well adapted to the

Dr. David P. environment, but we know that adaptation doesn't mean becoming more complex always. So sharks and crocodiles are

Dr. David P. are cited as species that haven't evolved much at all in many millions of years cavefish actually evolved to become more simple.

So basically fish with eyes got isolated in like an underground environment where there was no need for sunlight. And so basically fish like processing light like eyes didn't have any survival advantage. So basically fish lost their eyes over many generations of evolution becoming simpler. So if that's true why should evolution increase complexity?

The answer to that it comes from this understanding that evolutionary epistemology says that a species adapting to a niche is like a scientific theory being tested for viability. Species will evolve not to become increasingly complex but to match the complexity of the environment. A sort of simplest solution to the thermodynamic problem of staying far from equilibrium. So Einstein famously said a theory should be as simple as possible but no simpler. The same could be said about an organism. You want it to be as simple as possible while it accomplishes its task of extracting energy. It has to be complex enough to be able to extract energy in a competitive environment. But a bacterium doesn't have to do it doesn't need intelligence. Any extra information processing higher level cognition would be energy wasted that could be energy that allows it to stay far from equilibrium for longer

and replicate more. So it's not the case that everything will evolve towards higher complexity.

But the reason that we do get increasingly complex species over evolutionary time in a way that I'm saying is inevitable. So there is a progressive evolutionary process that leads to higher forms. It has to do with

a niche emergence. So basically a species will evolve to become as complex as its niche, as its environmental niche.

But there will be increasingly complex niches that emerge for reasons that will be explained. But I will pause there

before I get to that which will be what we end on.

There are no other questions right now. So if anyone does have questions, how about they can feel free to ask it during this last section, give you a second to catch your breath.

I know it's been a ton of information. Like, my goal is that, you know, since this is being recorded, it'll be something that's up. Hopefully, I've cited enough literature that people can explore this stuff themselves and see that like, okay, it's not this guy that's like,

I'm not going to have like, I'm not going to have like a theory of everything. This isn't my theory. It's a synthesis that's emerging. And really, the story has been there for quite some time. And the cybernetics pioneers, actually, a lot of them were you could call them like cosmic teleologists, a lot of them did see this continual increase in complexity.

Of course, there's nothing mystical about it or supernatural about it. It's a mechanistic process. That doesn't change the fact that nature seems to have these built in goals that emerge from the fundamental laws and constants, the parameters, the fine tuning. But it's a completely natural, progressive teleological evolutionary process.

And so I'll just end up, you know, this is just finishing up, it's along the same line. So that's good. Questions can come after this. I know, it's been quite a lot. So I will just try to sum this up simply. So two important principles from cybernetics. These came from Ross Ashby, who also created the principle of self organization, which was really the first time people took self organization, which was really the first time people took self organization.

So we think of complexity science, seriously. So we think of complexity science as being the science of self organization, but it really goes back to cybernetics. I think the term I think Kant might have talked about it, but it was Ross Ashby, who tried to, you know, give it a, make it a principle and a formalized concept.

So there are these two concepts that really, I think anybody interested in like Carl Fursten's Bayesian brain hypothesis free energy principle should be aware of, because all of the way he talks about like evolution comes from this Ashby's work, the good regulator theorem.

It was also updated. It was also updated. There was the internal model principle, which came like a decade later, good regulator theorem came.

He's both these laws, I think came in the fifties and the forties or fifties. And then I think in the seventies, the internal model principle was created, but it basically says that any system that regulates or controls another system must have a model of that system.

Otherwise it can't regulate it. So it seems pretty obvious, but this talk of models kind of starts here.

That, that is very familiar to like anyone interested in like the free energy principle.

So this, this can be applied to inanimate systems. I mean, controllers are like house, like thermostats and

stuff, but Ashby was really interested in adaptive systems.

So he was thinking about evolution. So applied to evolution, an organism must model its environment.

Uh, if it's going to extract free energy, uh, and an organism is a model of its niche, like a key is a model of the lock it opens.

So when we talk about these models can be kind of extract kind of abstract, but I think this model, a key is a model of the lock it opens really explains how an organism fits into a niche.

Uh, you could also say like a hand fits into a glove, but it really, uh, the organism is a model of its environment.

So natural selection is essentially an information channel that through evolution is pumping in information from the external world, from the environment into the organism, such that the organism is encoding the structure of reality and encoding the structure of the world.

And its design, uh, include, uh, uh, encodes information about the outside world.

So it's kind of a new way to think about evolution as this process of basically the universe is modeling itself and waking up through this process.

It's pretty interesting.

Um, kind of psychedelic.

So, um, the law of requisite variety is just telling you about the sophistication of that model.

So how complex does the model have to be for the organism to stay far from equilibrium?

Um, Ross Ashby's law says, so John non researcher at Cambridge, uh, kind of summed it up this way in colloquial terms.

Ashby's law has come to be understood as a simple proposition.

If a system is to be able to deal successfully with the diversity of challenges that its environment produces, then it needs to have a repertoire of responses, which is at least as nuanced as the problems thrown up by the environment.

So a viable system is one that can handle the variability of its environment, or as Ashby put it, only variety can absorb variety.

So it's what we said before with evolutionary epistemology, an organism evolves to become as complex as its environmental niche requires.

It has to specifically have enough behavioral responses, which map onto internal states.

So it has to have enough cognitive states to, uh, respond to the number of challenges that the environment presents challenges, um, uh, in its ability to extract energy.

And when, when, when you get organisms and you get other, uh, you get other agents, other organisms, then of course, what the organism has to model is much more complicated because it doesn't have to just model the energy source and its environment.

Like to get food, it also has to model other modelers, other cognitive agents.

Um, so yeah, basically any adaptive system has, uh, uh, uh, uh, a behavioral repertoire and a repertoire of accessible mental states or for things without brains, computational states.

And the number of states that the organism can access, uh, uh, uh, should match the number of challenges, the number of states that could surprise the organism by the environment.

Um, so a cat must have at least as many states as the ways the mouse can evade it and the mouse has to

have enough behavioral states to get away.

So they model each other, a swordsman, like a fencer must have many blocks as his opponent has attacks.

And as we said, because behavioral responses map onto unique internal states, the loss of an organism must have as many accessible states as required by the complexity of the niche.

So we already said this, uh, organism is some simplest solution to the thermodynamic problem of staying far from equilibrium.

So a well-adapted species represents biological solution to an existential thermodynamic dilemma.

It's a sort of living, evolving scientific theory about how to most efficiently extract the free energy, the lifeblood of existence out of a particular niche.

Some niches present a changing variety of challenges that must be adapted to while others present hardly any at all.

So almost done here.

I know it's a lot.

Hopefully this can kind of wrap up the ideas and kind of show how they're all connected.

Um, but this is still on this question of how complex is the universe?

Uh, how complex can it get?

So, uh, Olivia Judson wrote this, uh, uh, published this paper called the Energy Expansions of Evolution.

It's really nice.

People should check it out.

Uh, so she says the history of the life earth system can be divided into five energetic epochs, each featuring the evolution of life forms that can exploit a new source of energy.

These sources are geochemical energy, sunlight, oxygen, flesh, and fire.

The first two were present at the start, but oxygen, flesh, and fire are all consequences of evolutionary events.

And then she makes this statement, by the way, red is quotes if people haven't, uh, figured that out.

So these are her words.

I forgot to put quote marks.

Um, so no category of energy sources disappeared.

This has over time resulted in an expanding realm of the sources of energy available to living organisms and a concomitant, uh, increase in the diversity and complexity of ecosystems.

So here's the way to think about it.

Um, so taking the thermodynamic perspective and looking at organisms as energy flow channels, uh, we can think of each niche on earth as a sort of energy slot for a given species.

And an evolving population of organisms efficiently searching the solution space will eventually just by chance discover a solution to a thermodynamic problem that it didn't know existed.

The discovery of a novel energy source or energy extraction technique is both how a new niche and a new species come into existence.

So phylogenetic learning, which we saw in that chart will naturally lead to speciation because organisms will stumble upon new ways to exploit thermodynamic niches that were previously inaccessible to life purely for design reasons.

We're going to give examples on the next slide.

Um, so as the biosphere accumulates knowledge through phylogenetic learning, not only is the solution space corresponding to one kind of thermodynamic problem being explored, there's a growing problem space, each with its own unique solution space.

Um, so in an evolution, so there's an evolutionary trajectory that's not determined in the strict sense, uh, envisioned by Laplace where like there's no freedom in the future is like set in stone, but there is a sort of, uh, statistical determinism, what I've been calling a non-equilibrium statistical determinism that guarantees that basically the inevitability of this emergence of a distribution of far from equal equilibrium attractors.

sunlight was a little bit of a more difficult challenge to extract sunlight.

It took some time for evolution to be able to do that.

Um, so, uh, extracting energy from the hydrothermal vents, uh, source of energy is really simple because the energy is already like flowing through the system.

It doesn't have to do much work to extract sunlight.

So if you have, you know, a moving sun, uh, it's a little bit more complicated, but not nearly as complicated as organisms that eat other organisms.

So heterotrophic organisms, um, because they will have to model not just like a sun moving predictably through the air, but have to model all of these other agents.

agents, many of which are trying to eat them to kill you. So

basically, so answering this question, why increasingly intelligent species emerge, we started off with reductive autotrophs, those organisms around hydrothermal vents, they were a solution to the problem of how to extract free energy from geochemical gradients, photosynthetic bacteria, the ancestors of plants for the solution to extract of how to extract all the solar energy that was flowing through the planetary system. So that was a big discovery made by life is that they can get energy directly from sunlight.

Heterotrophic organisms, organisms that eat other organisms were the solution to the problem of how to extract energy from life itself. Once life starts having to model life, other agents with causal power and adaptive behavior, the computational task of extracting free energy gets increasingly difficult as increasingly complex species arise. So by virtue of having to model each other, the complexity of some species gets ratcheted up by what's known as an evolutionary arms race. So the lion is evolving, you know, lions are producing offspring, some of those lions will be better at predicting the gazelle's movements, the gazelle will also be creating offspring, and some of those will

be better at predicting the lion's attack. The ones that are better predicting are the ones that get to stick around. So you see how you have this ratcheting mechanism that ratchets up ratchets up complexity. There's actually a principle called the Red Queen principle that says that for an organism to simply stay in the game of existence, certain organisms, a certain species, they will have to become increasingly complex. And that comes from Alice in Wonderland. Basically, there's a scene with the Red Queen and Alice is there, they're all running as fast as they can. And Alice is like, you know, I'm running fast as I keep running, but I'm going nowhere. Like, so they're, they're all running, but they're not making any distance, they're staying in place. And the Queen says something like, Oh, you see here for that, to go somewhere, you'll have to run much faster than that. So the idea is that for a species that has to deal with this complex environment, just to stay where it is just to stay in the game of existence, it has to become increasingly complex, it has to keep adapting. So one thing that I wanted to get into more, but I'm not going to, is this idea of empowerment, it's comes from the computational neuroscience literature. It's similar, it comes from Daniel Polanyi, but it's very similar to Harvard's Alex Wisner, Visner Gross. He has an idea called causal and tropic forcing, but basically, these people are arguing that, well, Visner Gross is arguing that a causal and tropic force means that, like, basically, with like evolution, you'll get a system systems that are trying to maximize the number of states they can respond to. So the maximize and the number of states they can respond to is mirrored by their internal diversity. So systems naturally try to increase the number of states. And it's this ability that allows for intelligence, because if you have more states, going back to this law of requisite variety, you can respond to more challenges. So not every species is increasing empowerment, but you will get the emergence of increasing, increasingly complex species because each species serves as food for a potential new species. You will get increasingly complex species that emerge over time. And they will be increasingly empowered, meaning that they can respond to more environmental states, and that they have more accessible mental or cognitive states. I explained some of this in the last slide, but I'll just explain a little more. So open-ended niche emergence, why has the evolutionary records shown a trend of the emergence of increasingly complex forms? Because when all conceivable niches on earth were filled, that wasn't the end. New species created new niches because the free energy slot they provide is themselves, their food. So cybernetics guy, he's kind of like the last living cyberneticist, kind of. I'm sure there are other people that call themselves that, but he's kind of kept that title even when it like wasn't trendy. He says, it is well documented by evolutionary biologists that ecosystems tend to become more complex. The number of different species increases

and the number of dependencies and other linkages between species increases. He cites E. O. Wilson, who recently passed away, who did a lot of work that kind of inspired a lot of the concepts in the book. Not just this niche emergence, but also talking about super organisms, so collectives like ant colonies, self-organizing systems at all scales. But he said, not only do ecosystems contain typically lots of niches that will eventually be filled by new species, but there is a self-reinforcing tendency to create new niches. What does that mean? It means the biosphere, certainly ecosystems, are themselves autocatalytic sets. So Seth Lloyd, he's talking about these chemical autocatalytic sets, but it's, I should have had this on the earlier slide, but it'll be a good way to kind of show the equivalence between ecosystems.

Seth Lloyd says, autocatalytic sets of reactions are powerful systems. In addition to computing, they can produce a wide variety of chemical outputs. In effect, an autocatalytic set of reactions is like a tiny computer-controlled factory for producing chemicals. Some of these chemicals are the constituents of life. Now, Stuart Kaufman, big, you know, one of the main names who's kind of this whole talk, he's kind of put these ideas out there since like the early 90s. Well, actually going back farther than that, but he started writing books in the 90s and really gaining exposure for the Santa Fe Institute. He's written a recent paper, well, semi-recent, about Nietzsche emergence as an autocatalytic process. So ecosystems are autocatalytic, autocatalytically closed, self-sustaining reaction networks. So they're autopoietic agents that reliably drive up biological diversity and complexity as they self-amplify and evolve. So when you think about it, really all integrated networks of adaptive systems function as autocatalytic sets. There's been a lot of papers showing this. Economies, in a sense, are autocatalytic sets, including the social organisms we call societies or civilizations. So these networks are common because they emerge spontaneously when many organisms repeatedly interact with one another and discover

synergistic collective configurations as nature often pressures them to do. So we're at the end, almost. We've explained why complexity increases. Evolution will not increase the complexity of every species. They will become adapted to their niche. They will match the complexity of the niche. However, because new niches are always emerging, because species act as food for a potential new species, because of that process, you will get species that emerge that have a higher number of a larger repertoire of mental and behavioral states. So the law of requisite variety really explains why more complex environments create more complex niches. And this is open ended. There's really no limit to how far this process can go, and so especially since humans who are the pinnacle of this process, we're not, you know, a lot of people have been against these kinds of views because it seems like it's saying that humans are superior. We're superior maybe computationally, but not superior in any other sense. We're actually part of the biosphere, which is this integrated interconnected system. So it's really, you know, it doesn't make sense to think of us as like superior to nature because the whole biosphere is an organism that we're just sort of an organ for. We're basically the nervous system of the planet, the cybernetic global system that people have called Gaia. Some people don't like that name. I think it's a perfectly good name.

But that's why complexity gets ratcheted up. But the final,

I guess, part of this complexity increase story is that not only do you get increasingly complex species,

those species will interact naturally to form higher level adaptive systems which are collectives. So I call this the principle of recursive self-organization based on Ashby's principle of self-organization. What I'm arguing is that when you have a biosphere with these agents and you have some are simple and some are complex, naturally those agents will interact. When they interact, they will in many cases discover synergistic collective configurations. So configurations that make it easier for each agent to extract the energy they need to stay far from equilibrium. So why do interacting agents link up to form stable holes? For the exact same reason that molecules with the right chemical diversity will form stable autocatalytic reaction sets when pushed by a flow of energy. Working collectively allows the whole system to extract more free energy with less work. We can, that's the whole point of synergy is that working together makes your workload easier because there's a distribution of labor. And it's, it's really nice that, you know, this distribution of species with some simple and some complex, that's really what you need to create this like cybernetic adaptive system at the level of a planet because a system needs diversity among its components in structure and function for there to be this distribution of labor, labor, which makes it this uh, collective whole, this, this synergistic, uh, holistic unit. Um, so it's actually good that evolution and adaptation isn't increasing this, the complexity of each species. Like each species isn't becoming intelligent, like roaches aren't becoming like more and more intelligent over time with no limit. Um, the biosphere wouldn't be stable if that were the case. Uh, it's really, uh, ecosystems need this diversity among components just as a car with made of all engines isn't going to be functional or organism made of all brains. You need a variety of parts, some simple, some complex because they all do different functions. Um, so nature promotes cooperation, collaboration, and synergy because it is thermodynamically beneficial for all parties. Um, synergistic collective configurations will eventually be discovered by many component systems that are exploring various states or configurations through the blind variation and selective retention mechanism mentioned. Organisms only compete until they finally figure out what that working together makes everyone's task easier. And that goes for humans too. So it's kind of a moral lesson for everybody. The nations have to work together. We shouldn't, um, get rid of the nations. There shouldn't just be this evolution into this like global government because the nations are actually, uh, provide this diversity that we're saying is important, uh, because you need this, uh, diversity of component parts to create a division of labor. So there's strength in diversity. This is different than the evolutionary picture of survival of the fittest. Whereas there's just like this idea that the stronger, the most intelligent survive. This is saying that that's not true. It's the most adaptive that survive has nothing to do with strength or intelligence. Sloths have been around for many millions of years, despite being sloths being slow, uh, because they exploit a source of free energy. That's not being exploited by another, uh, species. So, um, going back to this kind of global message, you want nations to retain their identities, but you also want them at the same time to come together and to, uh, work synergistically. So you basically want something like a global system, but this, there has to be the optimal balance between centralization and decentralization. Um, neither approach will be good on its own. There needs to be this balance. So, um, as long as the biosphere is creating a growing variety of complex

adaptive systems, some of those will interact to produce higher level complex adaptive systems, which will come together to form even larger ones. This process continues, uh, at higher scales. So I didn't go into this, uh, left out a kind of a neat picture that shows a brain and then the planet and it's like information networks. But, uh, I do, I hope that, you know, talk about Gaia and global brain will become, uh, will be taken as seriously as it should be. Uh, they really only weren't taken seriously because there were mystical notions associated with this word Gaia, which wasn't even James, uh, uh, Lovelock's, uh, fault. He act, the name Gaia came from his neighbor who is actually the author of Lord of the Flies, William Golding. So blame him for the, for the flowery metaphor with Gaia. Um, but yeah, the, the, the biosphere is a cybernetic organism and we are forming something like the global brain and a self replicating biosphere, uh, is essentially the next step. And when we are trying to care form planets like Mars, that can be seen as the biosphere replicating, uh, for humans to at this point in our stage of development, we would have to kind of convert Mars to like a something like a biosphere to have like oxygen. And so you can see how it's replication with variation. You get the same sort of thing, but because it's a different planet, there's going to be a variation. It's not going to be exactly the same, but that's how the evolutionary process continues. I should have showed this earlier. This is just showing how like eukaryotic single celled organisms like amoebas will come together to form communities like a slime mold. Uh, these, uh, multi-cellular organisms come together to form colonies. You see this ant colony and actually really do form a super organism. You can see here, they actually build a bridge out of their bodies so that some ants can like climb across that bridge. So it really makes sense to call it a super organism or, uh, at least like a, a super adaptive system or like a meta, meta system. Um, because without it, we can't make sense of this collective behavior that's functional, um, of course, human civilization. We again, see this hierarchy. Now it makes more sense. Uh, we see that these evolutionary transitions. Oh yeah. These are called evolutionary transitions. Sometimes I use the name meta system transition because, um, if we were talking about the brain, I would also talk about like, uh, the emergence of a prefrontal cortex, which is like a high level controller. Um, that's also like a meta system transition. So meta system transitions are a little extend this where it's not talking necessarily about evolutionary transitions, but these other transitions where systems start to model themselves. Um, so here we see, uh, how evolution has created these different memory, uh, systems, these knowledge repositories. So it's that process of recursive self-organization that creates new memory systems. When you get go from single cell life to multicellular life to life with brains, uh, it's, uh, these are in, uh, revolutions in information storage machinery. So you can see how the information in the biosphere gets accumulated through an evolutionary process. Um, these are the final points. Yeah. So knowledge is powers, not just the hollow buzz phrase of the digital age. It's true in the most fundamental way. Uncertainty reducing information is life's first and last weapon in ongoing war with disorder. Without knowledge, life cannot exist for more than a moment, much less colonized the galaxy and beyond. This suggests that setting out on the path toward cosmic superiority is not a choice that intelligent agents like homo sapiens make upon careful reflection, nor is it just some quirky ambition we stumbled onto by chance. This is the main point humanity's collective desire to

transcend mortality and expand outward into space. So apparent from our current scientific and technological endeavors, SpaceX, uh, this doesn't emerge as an accident. This is an inevitable consequence of the fact that continual knowledge acquisition is a fundamental biological imperative. If life wants to persist and stay far from equilibrium, at some point it has to get off the planet because the sun is, uh, going to die. And, uh, it's these sorts of problems, these existential problems, which, uh, create progress because it's our awareness of this problem, which, uh, forces us to, uh, search the configuration space for the solutions that solve the problem. So as natural selection pumps information from the inanimate world into life, nature begins modeling itself, encoding its own structure, and the universe begins to wake up. We are the cosmos come alive, not metaphorically, but literally. This is Ray Kurzweil's, uh, image from his book, The Singularity is Near.

You can see that he sees this, he calls this the, the, the destiny of, of life in the universe. And he's not shy about using that word. Um, again, it's not determined in a strict sense of replace. This, uh, kind of a statistical determinism that creates these, uh, series of attractors. And, um, you've seen famous people, uh, so, you know, some people see that idea and they'll have been reading Brian Green and listening to all these reductionists and think that's not a scientific view. Well, uh, that's not true. And, um, I hope to see more of these people speaking out about it. So Christophe Koch in Confessions of a Romantic Reductionist says, the rise of sentient life within time's wide circuit was inevitable. Teilhard Deschardins, who really came up with this, don't have time to talk about him, French philosopher and Jesuit you should look into, is correct in his view that the islands within the universe, if not the whole cosmos, are evolving toward ever greater complexity and self knowledge. To be clear, he's not saying that earth had to bear life that, you know, primates had to walk the African glass lands. But he says, I do believe that the laws of physics overwhelmingly favored the emergence of consciousness. The universe is a work in progress. Such a belief evokes Jer, Jeremiah, I don't know how to say a word from many biologists and philosophers, but the evidence from cosmology, biology, and history is compelling. So there's some, uh, advanced blurbs for this book. Uh, if you want to pre-order it, um, it comes out in June. Uh, but every, uh, order that I get from now until then I'll count towards my first week sales total. So that will help me get on bestsellers list.

If you want to pre-order it now and you email me, um, I will send you a copy soon of this novel. I also wrote called Road to Omega. I was pretty busy in the last five years. Yeah. I went into a cave for like four or five years, but this, uh, it's a vehicle for the science and philosophy in the book. Kind of like an anti-Ain Rand

I shrugged maybe the opposite philosophy. Um, and the road to Omega substack kind of turns this philosophy into kind of an effort to, um, save the world with, uh, science epistemology and blockchain technology.

Uh, blockchains are part of this self-organization process. And that's the longest talk I've ever given.

That's the longest I ever talked at one time. Probably my wife will probably say that's not true, but, uh, I'm tired. So let's, uh, open it up to questions and thank you for your immense patience.

It's awesome to hear and think about it. So there are two questions in the chat. First is from Dean. Dean asks, how would we explain caring through a lens of free will or no free will? Would free will deniers say there's no caring when it appears that agents can or do fully care?

Dean asks, how would you explain the, you know, free will be the best of free will?

Dean asks, how would you explain the best of free will? Yeah. So great question. Um, unfortunately, I don't have much to say without saying the things that I was going to bring up in the next talk, really explaining what free will is. I guess I should just say right now, uh, kind of a brief explanation of what I mean by free will. So I've, I hope I've made the argument that agency emerges, uh, in adaptive systems that start off as biological organisms.

And, um, to the person that are self driving car thing, I, uh, uh, just to mention that real quick, I don't think they are agents, but I do think, uh, it's possible to create systems with agency, um, for reasons that I haven't gone into, um, uh, it seems important that the systems have a certain type of architecture.

So not like a standard computer as a von Neumann architecture, and it's not distributed or integrated in the way the brain is.

So it's not integrating much information. So these systems won't have causal power. They won't be agents.

So neuromorphic hardware has been a lot of people think that's going to allow machines that can think and perhaps even be conscious.

I think maybe that's the way to go. It's probably a lot more complicated than that. It's a step in the right direction.

So agency is real. It's due to information, gaining causal power in a system, and it starts steering the system.

So you'll see a bacterium performing chemotaxis. But does it make sense to say that the bacterium has free will when it's going this way and that,

just because it's not predictable in the way an inanimate system is?

So I would say that they have agency, but they don't have free will.

What free will is, and this is not my idea, other people have kind of been talking in this way.

Kevin Mitchell has a book, Neurogeneticist.

He's coming out with a book called Agents, which is all about agency and free will.

But the idea is that we are agents with causal power, but we're sort of,

where our responses like are basically programmed responses that are responses that occur automatically that are informed by this information that's been built up through evolution and adaptive learning. So a lot of the time we go through the day and we're not really thinking about things like driving or like waking up and like, you know, going to the fridge. We're not putting conscious thought into that. So that wouldn't seem to be free will. To me, you're kind of acting on autopilot.

So what free will is, is a higher level of control that emerges with something like a prefrontal cortex in a global workspace. Basically free will allow a prefrontal cortex allows an agent to override its instinctive and automatic behaviors. And so free will is something that we have, but only if we're exercising this higher level of control, it's associated with cognitive control or executive

control, effortful control. And basically, if you have a healthy functioning prefrontal cortex, you're going to basically the conscious mind is monitoring consciousness is a monitor. So it's, it's monitoring your behaviors. And if it detects that there's suboptimal behavior, for whatever reason, like it's just automatic kind of instinctual, let's say someone says something that makes us angry, and we respond immediately with like pushing them or something. That's not a wise move. And that may be may happen when we're fearful, because of the amygdala response, and we just have this automatic behavior. But free will would be when the conscious mind overrides that process. And the bet actually the free will studies that everybody cites, as saying that we don't have free will. He actually never said that he said free will is in this veto power, like maybe some of our voluntary movements occur without conscious thought. But it's our ability to override those movements. That is the source of free will. But there was more problems with that actually, the studies have coped to the study showing that basically, that free will, readiness potential, which is like a spike in electrical activity that occurs before you do this voluntary movement, this movement, you think is voluntary. And everyone thought that this meant that you're not really making a conscious decision that you could see this spike in brain activity earlier than the person thought they made the voluntary voluntary decision. But a new study has shown that that's only the case when we're making like arbitrary decisions. And when they had people making a decision about when when to donate like a large amount of money to a charity, you actually saw this readiness potential disappear. And so it seems like deliberate thinking, which is still this like prefrontal cortex. And there's this distinction between access, access consciousness and phenomenal consciousness that I didn't go into that I would in the next talk that Adam Saffron and integrated world model theory really gets at. But the idea is that the prefrontal cortex does allow us to override automatic behaviors. And that is a sort of freedom that we have, and we also lose it. In cases of schizophrenia, you see impaired prefrontal cortex activity and drug addicts, you see impaired activity. And drug addicts, you see, they do repetitive behavior, even when they know that it's not beneficial to them, they will still engage in that. So they're kind of stuck in this loop or this unhealthy attractor. With schizophrenia, there's patients that report being pushed around by forces that are beyond their control. I think that's because when the prefrontal cortex is basically deactivated, people lose this sense, this ability to override their automatic behaviors. And I think that's what those people are experiencing is sort of loss of this higher level agency that we've called free will. Another interesting, just a final interesting thing to mention about that is Cotard's syndrome, Cotard's delusion is this delusion where people think they're dead. They think they're like ghosts or just not alive. It's really weird. A lot of people die from starvation, because they think they don't have to eat because they think they're already dead. You actually see from neuroimaging studies, they've showed like impaired activity in the prefrontal cortex. So maybe these people who

think they're dead have lost this high level agency, and they feel like they're ghosts or something. As far as caring, caring is a part of like emerge through evolution, because you do have this benefit to cooperating with each other. This self-organization process that we talked about that happens naturally, because basically you find that when you work together, you minimize conflict and align it, you align with the same interests. And so caring, altruism, empathy, all of those things are as natural and as Darwinian as anything else. But you have to add on to evolutionary theory, all of this stuff about self-organization and cooperative evolution. As far as you say, he said something about why people who don't believe in free will. I forget the exact point about the caring. What was that?

When it appears that agents can or do fully care.

I'm sorry, what was the first part of it?

How would you explain caring through a lens of free will or no free will? Would free will deniers say there is no caring when it appears that agents can or do fully care?

Yeah, no, it flies in the face of our everyday experience. The amount of contradictions and logic that doesn't make sense when you go down that path of there being no free will or agency, you just get into absurd territory. I think when we look back, it's going to look quaint that for so long that we thought that everything was determined in a strict sense. And it's really a big issue because these people who believe that if they believe it, if they take it to heart, they're experiencing cognitive dissonance at every moment. And the people who argue against agency, they talk about religious people having to compartmentalize to deal with the real world. But anybody who believes that we don't have agency has to compartmentalize in the same way. Funny little conversation between Sam Harris and his wife, Annika Harris, who wrote a book called *Conscious* that is a great accessible book, but it leaves out everything that I've discussed and that I would discuss about the Bayesian brain hypothesis, global workspace theory doesn't really go into integrated information theory in any depth.

So like all of the things that they're missing that do explain a lot of this stuff can be found in like modern neuroscience. So we are making a lot of progress towards understanding the hard problem of consciousness, so subjective experience, but also a free will and agency. But it's funny because they're having this conversation in the podcast. And they're discussing whether since a belief in no free will, as mentioned that New York Times article, I think 2011 showed that they have less moral behavior, and it can, you know, cause depression, they're discussing whether they should tell their daughter about free will or not, that she doesn't have any free will. The discussion is like, should we tell our daughter that she doesn't have free will? But the actual discussion implies that they believe that they have some choice whether to tell her or not, they're having a discussion about whether they should do this or not. So there's a contradiction in their conversation that contradicts what they believe. What's interesting is that if they do tell her that she has no free will, and she believes it, she's actually gonna have negative mental effects, like it will, you know, be harmful, assuming that she really believes it. I don't know if it has that in everyone. But you do see studies that show that. So if we do have free will, it's a very bad thing to be telling people. And the last thing I'll say about that is, well, it's, it's interesting, because if she's familiar with epistemology, and they tell her that she doesn't

have free will, but she goes, Okay, that's based on a model that has uncertainty, and that model may be disproved, then she might not have the negative effects of free will. So I think everyone should practice being a good Bayesian, and understand that our models, our theories, even the one that I just talked about, will have errors, and will be shown to be wrong. And if reductionists do that, they won't call free will, belief in free will, like, comparing them to horoscopes, which is just, I think, a horrible statement. And kind of funny that physicists don't like it when like neuroscientists start, and biologists start like talking about physics. It's kind of strange that physicists are so sure that we don't have free will, when they're not familiar at all with the neuroscience behind agency and free will. So would you like me to just read a comment in the chat? Otherwise, I think we can kind of come to a close.

If there's anything, yeah, I can do one more question.

Yeah, well, Marco and Dave and others, thanks a lot for the great comments and commentary in the chat. I hope that Bobby, I hope you check the live chat replay. And Sean O'Connor wrote, thanks for the good and informative talk. Dr. Azarian mentioned the idea of evolutionary arms race. I was hoping to recommend that he consider if he hasn't, how an intraspecific evolutionary arms race may relate.

Oh, I'm not familiar with the intraspecific. So,

Dr. Azarian mentioned the idea of a game theory, red queen, not between two species, but within one species.

Dr. Azarian mentioned the idea of a evolutionary arms race there too. Yeah, certainly, because you're competing with other organisms. And John Maynard Smith, who I mentioned earlier, talking about like this causal power of information, biology, he was the person to bring von Neumann's game theory to biological evolution. So he did all, he basically showed that there is this, you know, game theory sort of process being played out. I think he, his examples use different species. But I think within species, you would see the sorts, same sort of thing. Of course, if it's not humans, you know, if you have other species, people are competing, organisms are competing, you will get increased complexity, but there are limits. So just because there's these evolutionary arm races, within and between species, there will still be a limit, because they have to model the environment and just modeling all these unnecessary variables

Dr. Azarian mentioned the same thing. So it will ratchet up complexity in species. But

I think at least with the planet Earth and our biosphere, that, that humans are this sort of leading envelope of complexity. And so it really just is open ended with the most complex species. So the most complex species is the one that will evolve toward higher complexity. And really, there might be a limit there, there are reasons to think there are limits to how far human biological intelligence can emerge. There's, I mean, can evolve towards there's something called a cephalization limit, which is like, how big our brain can get, because of reasons like having to do with like the skull. And so, but what's interesting is part of this process, this evolutionary processes, as Popper and newer people like Dennett have mentioned, continues with science and culture and technology. So humans really have open ended complexity, because

we're, we're augmenting ourselves with technology. So while biology, biological complexity may have limits, there's no limits.

[illegible]

of our minds and all of our devices are.

And for that reason, we need to be worried about privacy,

but we also need to not be scared of technology.

We just need to use technology to become more human.

So we're not transcending like humanity and biology.

Technology will allow us to become hyper-biological.

Very interesting note, perhaps to close it on.

Anytime you want the dot to, we can make it happen.

It can be any format.

We can invite anyone else on.

Really awesome and interesting to think about.

Do you want any last words?

If I come back to talk about free-run agency,

it'd be great to have like Kevin Mitchell

or maybe Eric Coel who's done a lot of work

on top-down causation and causal emergence

that was really influential on the book.

If you know people, you can Twitter, tag them

or email us and email them and CC us,

or just contact us and we'll do that on our side.

Thanks, it's been super fun.

Thanks, Daniel.

I really appreciate it.

I appreciate you reading a draft of the manuscript,

just like in a few days to give me some feedback

that actually helps clear up little mistakes in the book.

I heard you mention Marco.

Thanks to Marco and anybody else who's been watching

and chiming in with questions.

Really appreciate it.

Sorry for the information overload.

If we could make this recording,
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Maybe some legal stuff, Delta eight or, or.

Water.

For people.

Micro dose water.

Stay hydrated.

For people where it's legal.

Keep stay hydrated.